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Clustering

Unsupervised learning:
- Type of ML where the model is trained on unlabeled data.
- Unlike in supervised learning which has input-output pairs, it focuses on identifying patterns and structures within the data itself.
- The goal is to explore the data to find hidden relationships and groupings.

Clustering: Grouping similar points together based on features.
Dimensionality reduction: Reducing the # of features in a dataset while retaining as much information as possible.
Anomaly detection: Identifying the unusual data points (outliers) that differ significantly from the majority of the data.
Associative rule learning: Discovering interesting relationships between variables in large datasets.

Clustering

- Group a set of objects in such a way that objects in the same group (or cluster) are more similar to each other than to those in other groups.
- Similarity within clusters: Data points in the same cluster are more similar to each other.
- Dissimilarity between clusters: Data points in different clusters are more dissimilar.

Given a dataset $x = \{x_1, x_2, \ldots, x_n\}$, where $x_i \in$ P-dim space.
The goal is to partition the dataset into clusters $C = \{C_1, C_2, \ldots, C_k\}$

Hard clustering: Each data point belongs to a cluster completely or not.

Data points	F_1	F_2	cluster ($k=2$)
→	1	A	C_1
→	2	A	C_2
→	3	B	C_2

Soft clustering: Instead of assigning each data point to a cluster completely, a probability or likelihood of that point being in each cluster is evaluated.

Data points	F_1	F_2	clusters ($k=2$) C_1	C_2
→	1	A	0.9	0.1
→	2	A	0.4	0.6
→	3	B	0.05	0.95

Types of clustering algorithms

Centroid-based clustering (partitioning methods):
- These partition the dataset into k distinct clusters, where each cluster is represented by a centre point (called the representative point).
- The goal is to minimize the variance between each cluster.
- E.g. k-means, k-medoids.

Density-based clustering (model based):
- Group together data points that are closely packed together, marking as outlier points that lie alone in low-density regions.
- E.g. DBSCAN (density based spatial clustering of applications with noise).

Connectivity-based clustering (hierarchical clustering):
- It builds a tree-like structure (dendrogram) representing nested clusters by either merging smaller clusters into larger ones (agglomerative) or splitting larger clusters into smaller ones (divisive).

Distribution-based clustering:
- Assume that the data points are generated from a mixture of several underlying probability distributions.
- E.g. Gaussian mixture models.

k-means clustering

- Partition a dataset into k distinct, non-overlapping groups (clusters) where each data point belongs to the cluster with the nearest mean.

1. Inititalization:
   Choose k initial centroids randomly from the dataset.
2. Assignment step:
   Assign each data point to the nearest centroid based on a distance metric.
3. Update step:
   Calculate the new centroids by taking the mean of all the data points assigned to each centroid.
4. Repeat until convergence:
   Repeat steps 2 and 3 until convergence.

- centroids are not changing
- assignments not changing
- max # of iterations

Distance calculation:
The distance (Euclidean) between a datapoint x_i and centroid c_k: $d(x_i, c_k) = \sqrt{\sum_{j=1}^{p} (x_{ij} - c_{jk})}$

cluster assignment: $\text{cluster}(x_i) = \underset{k}{\text{argmin}}\, d(x_i, c_k)$

centroid update: $c_k = \frac{1}{N_k} \sum_{x_i \in C_k} x_i$

Data Point	Feature 1 (x_1)	Feature 2 (x_2)
1	1.0	2.0
2	1.5	1.8
3	5.0	8.0
4	8.0	8.0
5	1.0	0.6
6	9.0	11.0

Let's choose $k = 2$ and apply k-means algorithm with initially chosen centroids as data points 1 and 3.

→ $c_1 = (1.0, 2.0)$, $c_2 = (5.0, 8.0)$

	distances from $c_1 = (1.0, 2.0)$	from $c_2 = (5.0, 8.0)$	assignment
1 (1.0, 2.0)	0		c_1
2 (1.5, 1.8)	0.5385	7.119	c_1
3 (5.0, 8.0)		0	c_2
4 (8.0, 8.0)	9.2195	3	c_2
5 (1.0, 0.6)	1.4	8.4	c_1
6 (9.0, 11.0)			c_2

Now the clusters are

$\{(1.0, 2.0), (1.5, 1.8), (1.0, 0.6)\}$, $\{(5.0, 8.0), (8.0, 8.0), (9.0, 11.0)\}$

$c_1 = (1.166, 1.466)$ $c_2 = (7.33, 9)$

Question: K-Means Clustering

Consider the following statements regarding the K-Means clustering algorithm:

✓ 1. K-Means clustering requires the number of clusters k to be specified before running the algorithm.

2. K-Means uses Euclidean distance as the only metric for calculating the distance between data points and centroids.

✓ 3. The algorithm can converge to different solutions based on the initial placement of centroids.

4. K-Means clustering can be applied to categorical data without any preprocessing.

✓ 5. The main objective of K-Means is to minimize the within-cluster variance.

Question: K-Means Clustering

Consider the following dataset with four data points in a 2-dimensional space:

Data Point	Feature 1 (x)	Feature 2 (y)
A	1	2
B	2	3
C	6	5
D	7	8

1. **Initialization**: Assume we choose $k = 2$ clusters and randomly select initial centroids as follows:
   - Centroid 1 (C_1): (1, 2)
   - Centroid 2 (C_2): (6, 5)
2. **Assignment Step**: Calculate the distance of each data point from the centroids using Euclidean distance and assign each point to the nearest centroid.

Calculate the total sum of squared distances from each data point to its assigned centroid after the first assignment.

	distance from $c_1(1,2)$	$c_2(6,5)$	assigned centroid
A	0		c_1
B	$\sqrt{2} = 1.41$	5.8	c_1
C		0	c_2
D		$\sqrt{10} 3.162$	c_2

$0^2 + (\sqrt{2})^2 + 0^2 + (\sqrt{10})^2 = 12$

k-medoids clustering

- Medoid: A medoid is a point in the cluster from which sum of distances to other points is minimal.
- Instead of using mean/centroid as the centre point of the cluster, we use medoid as the reference point.

1. Inititalization:
   Choose k initial medoids randomly from the dataset.
2. Assignment step:
   For each data point, calculate the distance to each medoid and assign the cluster with the nearest medoid.
3. Update step:
   For each cluster, select the data point that minimizes the sum of distances to all other points in the same cluster to be the new medoid.
   While the cost decreases: For each medoid m, for each data point o which is not a medoid in the same cluster,
   > Swap m and o and compute the cost from o.
   > If the cost from o is more than m, undo the swap.
   (PAM: partitioning around clustering)
4. Repeat until convergence:
   Repeat steps 2 and 3 until convergence.

eg: Dataset:

Data Point	Feature 1 (x_1)	Feature 2 (x_2)
A	1	2
B	1	4
C	1	0
D	10	2
E	10	4

Use k-mediods algorithm with k=2 with initial mediods as data points A and D. Use Euclidian distance as the distance metric.

→

Data points	Distance from A (1,2)	Distance from D (10,2)	cluster
(1,2)	0	9	C_1
(1,4)	2	9.2	C_1
(1,0)	2		C_1
(10,2)		0	C_2
(10,4)		2	C_2

$C_1 = \{(1,2), (1,4), (1,0)\}$ (A, B, C)

sum of distances from other points: $0+2+2=4$ ✓, 6, 6

$C_2 = \{(10,2), (10,4)\}$ (D, E)

2 ✓, 2

new medoids $\{(1,2), (10,2)\}$ medoids don't change.

clusters: $\{A, B, C\}, \{D, E\}$

eg: Consider the following dataset:

Data Point	Feature 1 (x_1)	Feature 2 (x_2)
1	1	2
2	2	~~2~~
3	3	4
4	6	2 4
5	8	3

Assume that you run K-mediods algorithm with k=2 and Data points 1 and 3 as the initial mediods using manhattan distance to compute

After the second assignment step, calculate the total sum of distances for all data points from their med

	d from (1,2)	d from (3,4)	cluster
(1,2)	0	4	m_1
(2,2)	1	3	m_1
(3,4)	4	0	m_2
(6,4)	7	3	m_2
(8,3)	8	6	m_2

clusters → {(1 2), (3 4 5)}

medoid = 1, medoid = 4

	m_1 (1,2)	m_2 (6,4)	
(1,2)	0	7	m_1
(2,2)	1	6	m_1
(3,4)	4	3	m_2
(6,4)	7	0	m_2
(8,3)	8	3	m_2

sum of distances from respective medoids

$= 0+1+3+0+3$

$= 7$

Hierarchical clustering

Unsupervised ML technique used to group similar data points into a hierarchy of clusters.
into a hierarchy of clusters.
It creates a tree-like structure (dendrogram) that represents nested clusters.

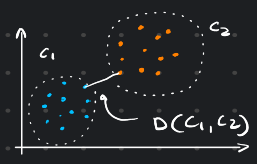
- Bottom-up approach (agglomerative hierarchical clustering):
  It starts with each data point as an individual cluster and iteratively merges the closest clusters until only 1 cluster remains or until a specified # of clusters is achieved.
- Top-down approach: It starts with 1 large cluster containing all data points and iteratively splits the clusters into smaller clusters until each data point is it's own cluster or until a specified condition is met.

Linkage criteria

- The distance between 2 clusters is crucial for hierarchical clustering. There are various ways to calculate the distance between 2 classes. These ways decide the rule for clustering. These measures are called linkage criteria/ methods.

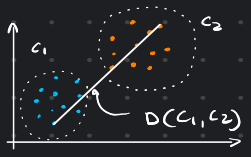
- Single linkage: (default)
  It is the distance between the closest points of the 2 clusters.

$$D(A,B) = \min\{d(A_i, B_j)\}$$



- Complete linkage:
  It is the distance between the farthest points of the 2 clusters.

$$D(A,B) = \max\{d(A_i, B_j)\}$$

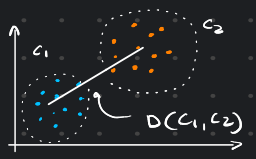


- Average linkage:
  It is the average distance between every pair of points in the 2 clusters.

$$D(A,B) = \frac{1}{|A|\,|B|} \sum_{a \in A} \sum_{b \in B} d(a,b)$$

- Centroid linkage:
  It is the distance between the centroids of the 2 clusters.

$$D(A,B) = d(\bar{a}, \bar{b})$$



e.g. $A: \{(1,2), (1,4)\}$ $B: \{(2,3), (5,7)\}$

$d(A_1, B_1) = \sqrt{1^2+1^2} = \sqrt{2}$ $d(A_2, B_1) = \sqrt{2}$ $d(A_1, B_2) = \sqrt{41}$ $d(A_2, B_2) = \sqrt{25}$

single linkage distance $= \sqrt{2}$
complete linkage distance $= \sqrt{41}$
average linkage distance $= \frac{1}{4}(\sqrt{2}+\sqrt{2}+\sqrt{41}+\sqrt{25}) = 3.5578$
centroid linkage distance:

$\bar{a} = (1,3)$ $\bar{b} = (3.5, 5)$

$d(\bar{a}, \bar{b}) = \sqrt{2.5^2 + 2^2} = 3.2015$

- Ward's linkage:
  It minimizes the total within-cluster variance when merging 2 clusters.
  It tends to create clusters of small size.





$$W(A) = \sum_{x \in A} \|x - \mu_A\|^2$$

$$W(B) = \sum_{x \in B} \|x - \mu_B\|^2$$

$$W(C) = \sum_{x \in A \cup B} \|x - \mu_C\|^2$$

Ward's distance btw 2 clusters is the increase in the variance when merging clusters A and B.

$$\Delta W = W(C) - [W(A) + W(B)]$$

$$= \frac{|A| \cdot |B|}{|A|+|B|} \cdot \|\mu_A - \mu_B\|^2$$

$\|\mu_A - \mu_B\|$: distance btw centroids of A and B.

Proof: $\mu_C = \frac{|A| \cdot \mu_A + |B| \cdot \mu_B}{|A| + |B|}$

$$w_C = \sum_{x \in A \cup B} \|x - \mu_C\|^2 = \sum_{x \in A} \|x - \mu_C\|^2 + \sum_{x \in B} \|x - \mu_C\|^2$$

$$\sum_{x \in A} \|x - \mu_C\|^2 = \sum_{x \in A} \left\| x - \frac{|A| \cdot \mu_A + |B| \mu_B}{|A| + |B|} \right\|^2$$

$$= \sum_{x \in A} \left\| (x - \mu_A) + \frac{|B|}{|A| + |B|} \cdot (\mu_B - \mu_A) \right\|^2$$

$$= \sum_{x \in A} \|(x - \mu_A)\|^2 + 2 \frac{|B|}{|A| + |B|} (\mu_B - \mu_A) \underbrace{\sum_{x \in A} (x - \mu_A)}_{0} + \left\| \frac{|B|}{|A| + |B|} \cdot (\mu_B - \mu_A) \right\|^2 \sum_{x \in A} \cdot 1$$

$$= w(A) + 0 + \frac{|A| |B|^2}{(|A| + |B|)^2} \|\mu_B - \mu_A\|^2$$

By symmetry

$$\sum_{x \in B} \|x - \mu_C\|^2 = w(B) + \frac{|B| \cdot |A|^2}{(|A| + |B|)^2} \|\mu_A - \mu_B\|^2$$

$$w(C) = w(A) + w(B) + \frac{|A| |B|^2 + |B| |A|^2}{(|A| + |B|)^2} \|\mu_A - \mu_B\|^2$$

$$w(C) = w(A) + w(B) + \frac{|A| |B|}{|A| + |B|} \|\mu_A - \mu_B\|^2$$

$$\Rightarrow \quad \Delta w = \frac{|A| \cdot |B|}{|A| + |B|} \cdot \|\mu_A - \mu_B\|^2$$

$A: \{(1,2), (1,4)\}$ $B: \{(2,3), (5,7)\}$

centroids: $(1, 3)$ $(3.5, 5)$

$$\Delta w = \frac{2 \cdot 2}{2+2} \left(\sqrt{2.5^2 + 2^2}\right)^2 = 10.25$$

Agglomerative hierarchical clustering

(bottom up)

1. Initialization: Each data point is treated as a separate cluster. For n data points, there are n clusters at the beginning.
2. Distance calculation: Calculate the pairwise distances between all clusters using a chosen distance metric and linkage criterion.
3. Merge clusters: Identify the two closest clusters based on the distance matrix and merge them into a single cluster.
4. Update the distance matrix: After merging, update the distance matrix to reflect the distances between the new cluster and the remaining clusters.
5. Repeat: Repeat steps 3, 4 until only 1 cluster is remaining (or till some stopping criterion is reached).

Eg: Consider a small dataset of four data points:

Data Point	Feature 1 (x1)	Feature 2 (x2)
A	1	2
B	1	4
C	2	2
D	4	3

Perform Agglomerative Hierarchical clustering using Euclidean distance and single linkage until single cluster is left.

→ A(1,2), B(1,4), C(2,2), D(4,3).

Distance matrix:

	A	B	C	D
A	0	$\sqrt{4}$	$\sqrt{1}$	$\sqrt{10}$
B	$\sqrt{4}$	0	$\sqrt{5}$	$\sqrt{10}$
C	$\sqrt{1}$	$\sqrt{5}$	0	$\sqrt{5}$
D	$\sqrt{10}$	$\sqrt{10}$	$\sqrt{5}$	0

Smallest distance = d(A,C) = 1
merge {A, C}
now clusters are {A,C}, {B}, {D}

Distance matrix:

	{A,C}	B	D
{A,C}	0	$\sqrt{4}$	$\sqrt{5}$
B	$\sqrt{4}$	0	$\sqrt{10}$
D	$\sqrt{5}$	$\sqrt{10}$	0

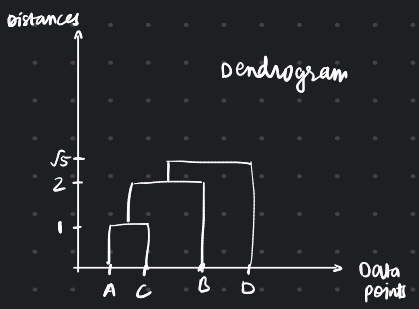
merge {A,C} and {B}

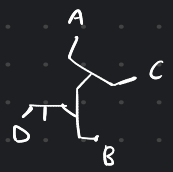
Distance matrix

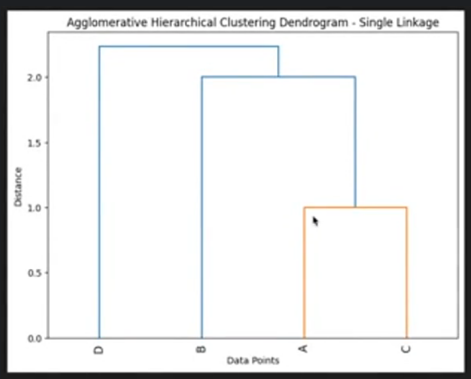
	{A,C,B}	D
{A,C,B}	0	$\sqrt{5}$
D	$\sqrt{5}$	0

merge {A,C,B} and {D}

clusters: {A,C,B,D}

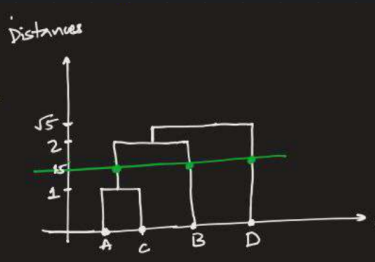






The dendrogram is a tree-like structure that is used to store each step of hierarchical clustering. Leaves of the dendrogram represent data points or observations in the dataset. Branches connect the leaves and represent the distance or dissimilarity between clusters.
The height at which 2 clusters are joined reflects the distance between them. A shorter branch means a smaller distance.

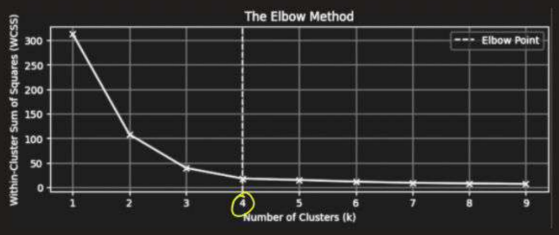
Cutting the dendrogram:
A horizontal line can be drawn across the dendrogram at a certain height to get a certain # of clusters.



↑ height of cutoff
↑ size of clusters
↓ number of clusters

Elbow method (optimal value of k):
For different values of k (# of clusters), plot the within cluster sum of squares, then identify the elbow point in the curve to find the optimal value of k.

K ↑ WCSS ↓

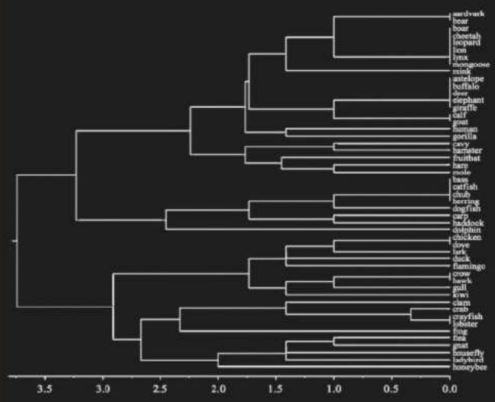


Before the elbow:
Increase in K significantly reduces WCSS.

After the elbow:
Adding more clusters results in a minimal reduction in WCSS.

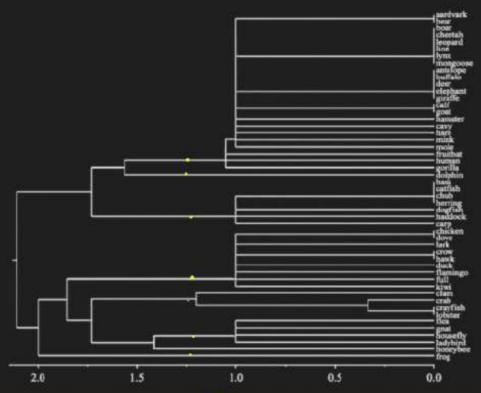
single linkage
↳ distance b/w clusters & data points is similar

Complete linkage
↳ distance b/w clusters ↑ data points ↓



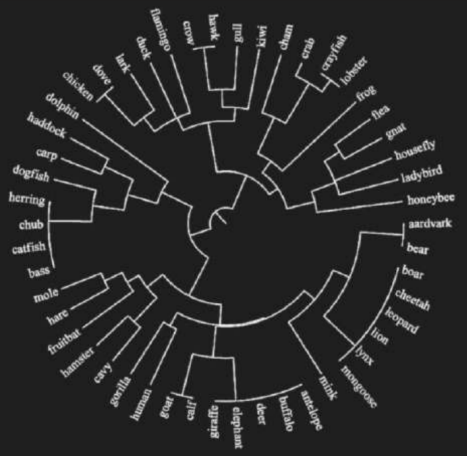
①

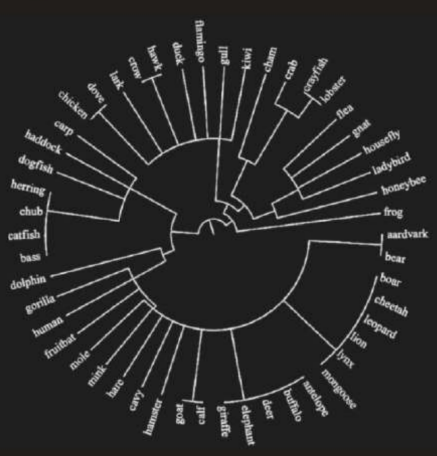
Complete Linkage
produces compact clusters at lower levels of the tree



②

Single linkage
produces clusters with large diameters at low levels of tree





Eg: Consider the following dataset in 2-D space:

Point	Feature 1	Feature 2
A	1	2
B	2	3
C	3	3
D	5	8
E	8	9

Using Complete linkage and manhattan distance as distance metric, perform AHC and draw the dendrogram.